



Shore School

2015

Year 11

Preliminary Task 5

Yearly Examination

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board approved calculators may be used
- Answer Questions 1 – 10 on the Multiple Choice answer sheet provided
- Start each of Questions 11 – 14 in a new writing booklet
- In Questions 11–14, show relevant mathematical reasoning and/or calculations
- Write your examination number on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A”

Note: Any time you have remaining should be spent revising your answers.

Examination Number:

Set:

Total marks – 70

Section I

Pages 2–5

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II

Pages 6–11

60 marks

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 What is the simplified expression of $(x-3)^2 - 4x(x+5)$?

- (A) $-3x^2 + 20x - 9$
- (B) $-3x^2 + 14x + 9$
- (C) $-3x^2 - 26x + 9$
- (D) $-3x^2 - 20x - 9$

2 A watch sells for \$148.00, including 10% GST. What is the price before GST is added?

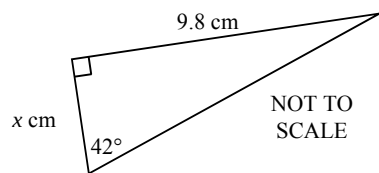
- (A) \$13.45
- (B) \$14.80
- (C) \$133.30
- (D) \$134.55

3 What is the factorised form of $x^3 - 8$?

- (A) $(x-2)(x^2 + 2x + 4)$
- (B) $(x-2)(x^2 - 2x + 4)$
- (C) $(x-2)(x^2 + 4x + 4)$
- (D) $(x-2)(x^2 - 4x + 4)$

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

- 4 What is the correct expression for the value of x ?



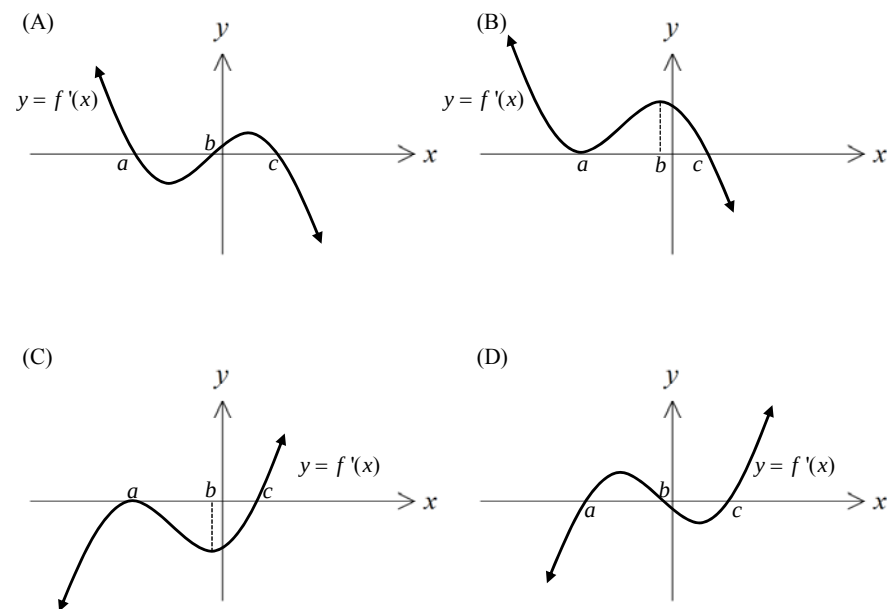
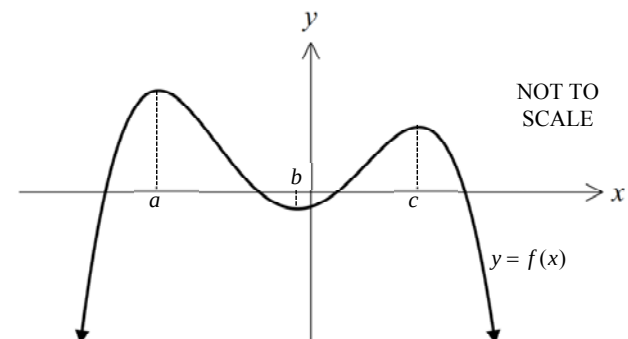
- (A) $x = \frac{9.8}{\tan 42^\circ}$
 (B) $x = 9.8 \tan 42^\circ$
 (C) $x = \frac{9.8}{\sin 42^\circ}$
 (D) $x = 9.8 \sin 42^\circ$
- 5 What is the gradient of the tangent to the curve $y = 4x^2$ at the point where $x = 3$?

- (A) -24
 (B) $-\frac{1}{24}$
 (C) 24
 (D) $\frac{1}{24}$

- 6 What is the perpendicular distance between the point $(-3, 7)$ and the line $2x - y + 5 = 0$?

- (A) $4\sqrt{3}$
 (B) $\frac{8\sqrt{5}}{5}$
 (C) $\frac{8\sqrt{3}}{3}$
 (D) $\frac{4\sqrt{58}}{29}$

- 7 What is the graph of the gradient function for the curve $y = f(x)$?



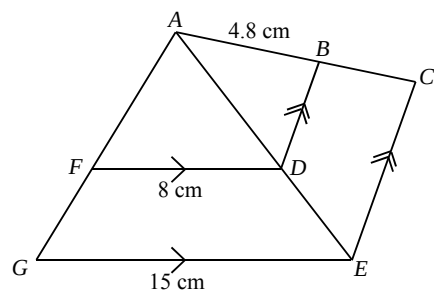
- 8 For what values of k will $x^2 + (k + 2)x + 4 = 0$ have no real roots?

- (A) $-6 \leq k \leq 2$
 (B) $-6 < k < 2$
 (C) $k \leq -6, k \geq 2$
 (D) $k < -6, k > 2$

- 9 How many solutions of the equation $\sin(2x) = 0.5$ lie between 0° and 360° ?

(A) 1
(B) 2
(C) 3
(D) 4

- 10 In the diagram $FD = 8$ cm, $GE = 15$ cm and $AB = 4.8$ cm.
What is the length of BC ?



(A) 2.6 cm
(B) 4.2 cm
(C) 7.2 cm
(D) 9.0 cm

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet

(a) Evaluate $\sqrt[3]{\frac{7\pi^2}{6}}$ correct to 3 significant figures. 2

(b) Simplify $\frac{x^2 - 2x - 8}{4y - 12} \times \frac{y^2 - 9}{6x - 24}$. 2

(c) Solve $|x - 8| = 3x - 4$. 3

(d) Express $\frac{5}{\sqrt{3} - 1}$ in simplified form with a rational denominator. 2

(e) State the domain and range for $y = \sqrt{49 - x^2}$. 2

(f) The function $g(x)$ is defined as: 2

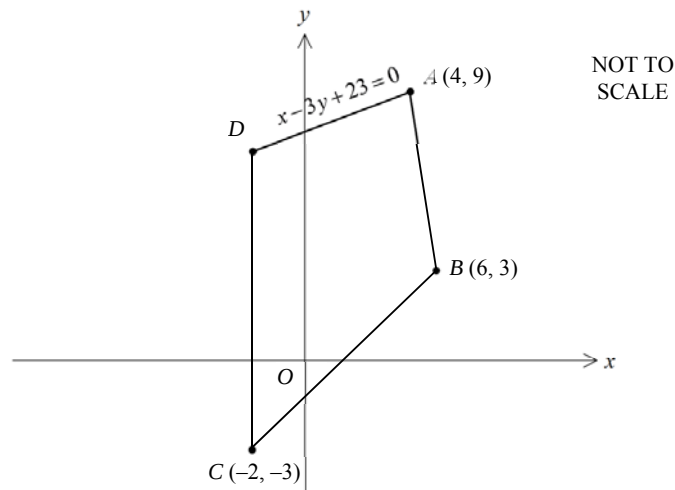
$$g(x) = \begin{cases} x^2 & \text{for } -5 \leq x \leq 0 \\ x - 3 & \text{for } 0 < x < 2 \\ 5 & \text{for } x \geq 2 \end{cases}$$

Evaluate $g(-3) - g(2) + g(1)$.

(g) Differentiate $\frac{2x - 1}{3x + 2}$. 2

Question 12 (15 marks) Use a SEPARATE writing booklet

- (a) $ABCD$ is a kite with vertices $A(4, 9)$, $B(6, 3)$, $C(-2, -3)$ and D .
The line AD has equation $x - 3y + 23 = 0$.

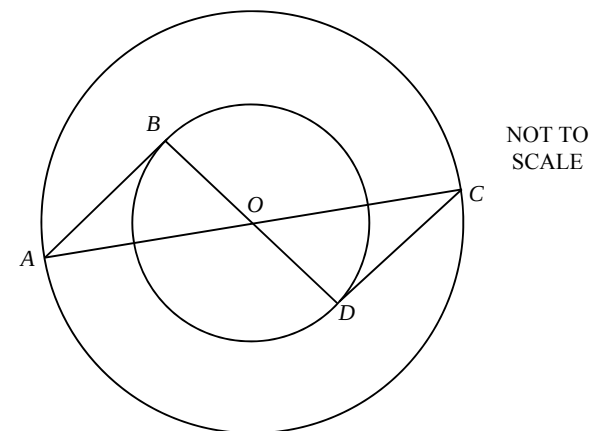


- | | |
|---|---|
| (i) Find the distance BC . | 1 |
| (ii) Show the equation of diagonal AC is $2x - y + 1 = 0$ | 2 |
| (iii) Given the line CD is parallel to the y -axis, find the coordinates of point D . | 1 |
| (iv) Show that the diagonals of the kite are perpendicular. | 1 |
- (b) Solve $\sqrt{3} \tan \theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$.
- (c) The roots of the quadratic equation $3x^2 + 6x + k = 0$ are α and β .
- | | |
|---|---|
| (i) Find the value of $\alpha + \beta$. | 1 |
| (ii) Find the value of k if $\alpha^2 + \beta^2 = -6$. | 3 |

Question 12 continues on page 8

Question 12 (continued)

- (d) Two circles have the same centre O . Lines AC and BD intersect at O .



- | | |
|--|---|
| (i) Prove $\triangle AOB \equiv \triangle COD$. | 2 |
| (ii) Hence, or otherwise, show $AB \parallel CD$. | 2 |

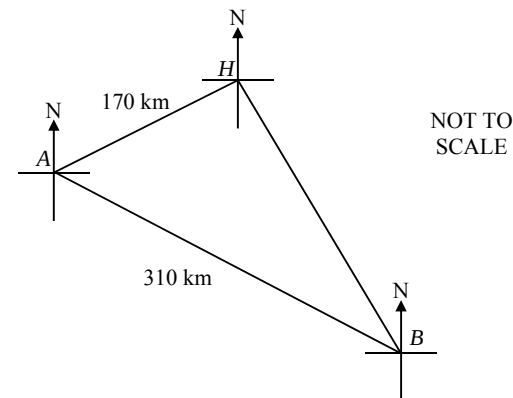
End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet

- (a) A function is defined by $f(x) = x^2 + 5x$.
- (i) Show that $f(x+h) = x^2 + 2xh + h^2 + 5x + 5h$. 1
- (ii) Hence, use the formula $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to differentiate $f(x) = x^2 + 5x$ from first principles. 2
- (b) Consider the two equations $y = -x^2$ and $x + y + 6 = 0$.
- (i) Solve the two equations simultaneously to find their points of intersection. 2
- (ii) Sketch $y = -x^2$ and $x + y + 6 = 0$ on the same number plane, labelling their points of intersection. 2
- (c) Shade the region represented by $x^2 + y^2 \leq 25$ and $x < 3$. Do not find their points of intersection. 2
- (d) Differentiate $f(x) = 5x^2(2x-1)^3$ leaving your answer in fully factorised form. 3
- (e) Find the equation of the normal to the curve $y = x^3 + 5x^2 - 4$ at the point where $x = 1$. 3

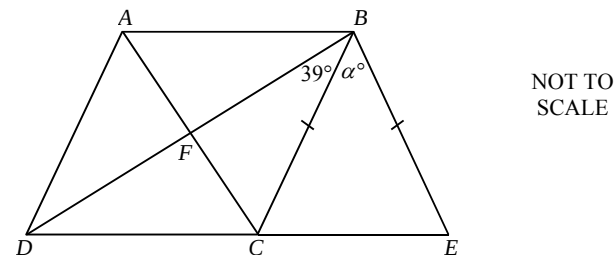
Question 14 (15 marks) Use a SEPARATE writing booklet

- (a) Ship A and Ship B leave the same harbour, H. Ship A sails on a bearing of 254° for 170 km. Ship B sails on a bearing of 117° until it is 310 km from ship A.



Copy or trace this diagram into your writing booklet.

- (i) Show that $\angle AHB = 137^\circ$. 1
- (ii) Hence, or otherwise, find the bearing of Ship A from Ship B. Give your answer correct to the nearest degree. 3
- (b) Prove $\frac{(1 - \cos \theta)(1 + \cos \theta)}{\cos^2 \theta} = \tan^2 \theta$. 2
- (c) ABCD is a rhombus, with DC produced to E. $\angle FBC = 39^\circ$ and $BC = BE$. 3



Find the value of α giving reasons.

Question 14 continues on page 11

Question 14 (continued)

- (d) Solve $9^x - 6(3^x) - 27 = 0$. **3**
- (e) A curve has equation $y = kx^3 + 2$, where k is a constant. At the point where $x = 2$ the tangent to the curve is inclined at an angle of 60° with the positive direction of the x -axis. Find the exact value of k . **3**

END OF PAPER

MULTI CHOICE

- ① C
② D
③ A
④ A
⑤ C

- ⑥ B
⑦ A
⑧ B
⑨ D
⑩ B

① $(x-3)^2 - 4x(x+5) = x^2 - 6x + 9 - 4x^2 - 20x$
 $= -3x^2 - 26x + 9$ ③

② $x \times 1.1 = 148.00$
 $x = 134.55$ ⑤

③ $x^3 - 8 = (x-2)(x^2 + 2x + 4)$ ①

④ $\tan 42 = \frac{9.8}{x}$
 $x = \frac{9.8}{\tan 42}$ ①

⑤ $y = 4x^2$
 $\frac{dy}{dx} = 8x$
 at $x=3$ $m = 8 \times 3 = 24$ ③

⑥ $(-3, 7)$ $2x - y + 5 = 0$

$d = \frac{|2x-3+(-4)+5|}{\sqrt{2^2+(-1)^2}}$
 $= \frac{|-8+1|}{\sqrt{5}}$
 $= \frac{8}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}}$
 $= \frac{8\sqrt{5}}{5}$ ⑧

⑧ $x^2 + (k+2)x + 4 = 0$
 $\Delta < 0$

⑨ $(k+2)^2 - 4 \times 1 \times 4 < 0$
 $k^2 + 4k + 4 - 16 < 0$
 $k^2 + 4k - 12 < 0$

critical points inequality.
 $(k+6)(k-2) = 0$
 $k = -6, k = 2$
 $\frac{x}{-6} \quad \frac{x}{2}$

$-6 < k < 2$ ⑥

⑩ $\sin(2x) = 0.5$ $0 \leq 2x \leq 720$
 $2x = \sin^{-1}(0.5)$
 $x = 30, 150, 390, 510$
 $x = 15, 75, 195, 255$

⑪ 4 solutions ⑤

⑫ $\frac{4.8+8c}{4.8} = \frac{15}{8}$
 $4.8+8c = 9$
 $8c = 4.2$ ⑧

Question 11

a) $\sqrt[3]{\frac{7n^2}{6}} = 2.258129...$
 $= 2.26$

b) $\frac{x^2 - 2x - 8}{4y - 12} \times \frac{y^2 - 9}{6x - 24} = \frac{(x-4)(x+2)}{4(y-3)} \times \frac{(y-3)(y+3)}{6(x-4)}$
 $= \frac{(x+2)(y+3)}{24}$

c) $|x-8| = 3x-4$

$x-8 = 3x-4$
 $-4 = 2x$
 $x = -2$
 $x-8 = -(3x-4)$
 $x-8 = -3x+4$
 $4x = 12$
 $x = 3$

test LHS = $|-2-8| = 10$
 RHS = $3 \times -2 - 4 = -10$
 $\therefore$ not a solution
 test LHS = $|3-8| = 5$
 RHS = $3 \times 3 - 4 = 5$
 $\therefore$ is a solution

d) $\frac{5}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = \frac{5(\sqrt{3}+1)}{(\sqrt{3})^2 - 1^2}$
 $= \frac{5(\sqrt{3}+1)}{2}$

e) $y = \sqrt{49 - x^2}$

domain $x: -7 \leq x \leq 7$
 range $y: 0 \leq y \leq 7$

$$f) g(x) = \begin{cases} x^2 & -5 \leq x \leq 0 \\ x-3 & 0 < x < 2 \\ 5 & x \geq 2 \end{cases}$$

$$g(-3) - g(2) + g(1) = (-3)^2 - 5 + (1-3) = 9 - 5 - 2 = 2$$

$$g) \frac{d}{dx} \frac{2x-1}{3x+2} = \frac{VU' - UV'}{V^2} \quad \begin{matrix} U = 2x-1 & V = 3x+2 \\ U' = 2 & V' = 3 \end{matrix}$$

$$= \frac{2(3x+2) - 3(2x-1)}{(3x+2)^2}$$

$$= \frac{6x+4-6x+3}{(3x+2)^2}$$

$$= \frac{7}{(3x+2)^2}$$

QUESTION 12

$$a) i) d = \sqrt{(3+3)^2 + (6+2)^2}$$

$$= \sqrt{100}$$

$$= 10$$

$$ii) M_{AC} = \frac{9+3}{4+2}$$

$$= \frac{12}{6}$$

$$= 2 \quad (4, 9)$$

$$y-9 = 2(x-4)$$

$$y-9 = 2x-8$$

$$0 = 2x - y + 1$$

$$iii) \text{ x-value of } D = -2$$

$$\therefore x-3y+23=0$$

$$-2-3y+23=0$$

$$21=3y$$

$$y=7$$

$$\therefore D(-2, 7)$$

$$iv) M_{AC} = 2$$

$$M_{BD} = \frac{3-7}{-2}$$

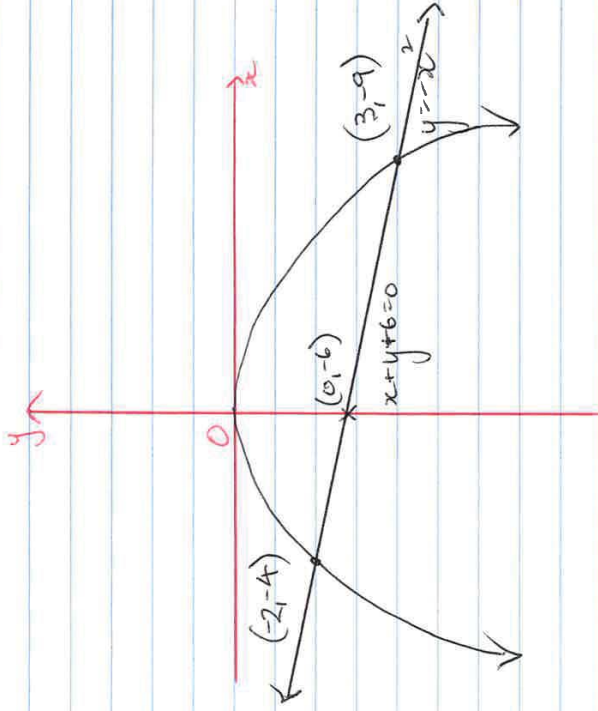
$$= \frac{-4}{-2}$$

$$= 2$$

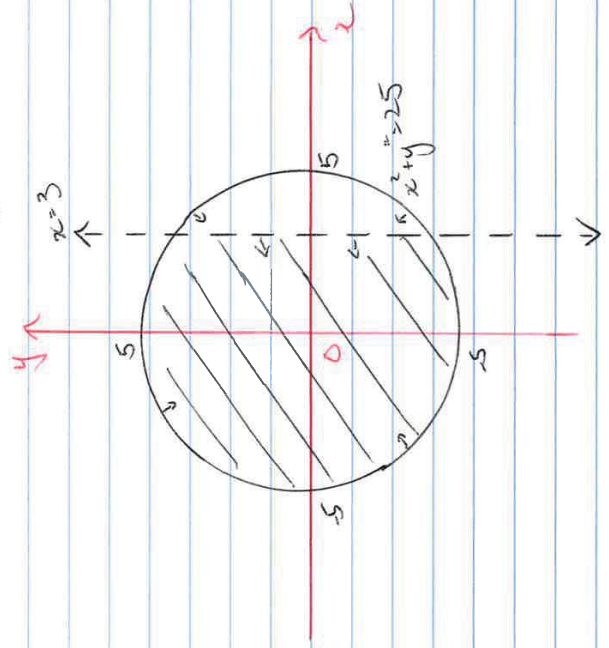
$$M_{AC} \times M_{BD} = 2 \times -\frac{1}{2}$$

$$= -1$$

$\therefore$ perpendicular



c) $x^2 + y^2 = 25$ (circle at $(0,0)$ with radius 5)
 $x=3$ (vertical line at $x=3$)



d) $f(x) = 5x^2(2x-1)^3$

$$u = 5x^2$$

$$u' = 10x$$

$$v = (2x-1)^3$$

$$v' = 3(2x-1)^2 \times 2$$

$$= 6(2x-1)^2$$

$$f'(x) = uv' + u'v$$

$$= 10x(2x-1)^3 + 5x^2 \times 6(2x-1)^2$$

$$= 10x(2x-1)^3 + 30x^2(2x-1)^2$$

$$= 10x(2x-1)^2 [2x-1 + 3x]$$

$$= 10x(2x-1)^2 (5x-1)$$

e) $y = x^3 + 5x^2 - 4$
 $y' = 3x^2 + 10x$

at $x=1$ at $x=1$ $y = 1^3 + 5 \times 1^2 - 4$
 $M = 3 \times 1^2 + 10 \times 1$
 $= 13$
 $M_L = -\frac{1}{13}$

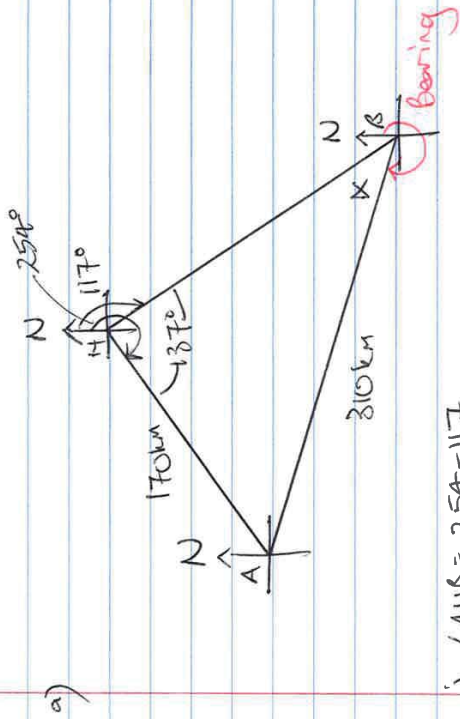
$$\therefore y - y_1 = m(x - x_1)$$

$$y - 2 = -\frac{1}{13}(x - 1)$$

$$-13y + 26 = x - 1$$

$$0 = x + 13y - 27$$

QUESTION 14



i) $\angle AHB = 254 - 117$
 $= 137^\circ$

ii) $\frac{\sin x}{170} = \frac{\sin 137}{310}$

$\sin x = \frac{170 \sin 137}{310}$

$x = \sin^{-1} \left(\frac{170 \sin 137}{310} \right)$

$x = 21.962 \dots$

$x = 22^\circ$

co-interior $\angle$ s.

$\therefore \text{Bearing} = 360 - 22 - (180 - 117)$
 $= 360 - 22 - 63$
 $= 275^\circ$

b) $\frac{(1 - \cos \theta)(1 + \cos \theta)}{\cos^2 \theta} = \tan^2 \theta$

LHS = $\frac{(1 - \cos \theta)(1 + \cos \theta)}{\cos^2 \theta}$

= $\frac{1 - \cos^2 \theta}{\cos^2 \theta}$

= $\frac{\sin^2 \theta}{\cos^2 \theta}$

= $\tan^2 \theta$
 = RHS

c) $\angle ABD = 39^\circ$ (diagonals of a rhombus bisect $\angle$ s)
 $\angle APC = 39 + 39$ (opposite $\angle$ s in a rhombus are equal)
 $= 78^\circ$

$\angle BCE = \angle ADC$ (corresponding $\angle$ s, $AB \parallel DC$)
 $= 78^\circ$

$\angle BEC = \angle BCE$ (base $\angle$ s of isosceles Δ are equal)
 $= 78^\circ$

$\therefore x = 180 - 78 - 78$ ($\angle$ sum Δ is 180°)
 $= 24^\circ$

d) $x^2 - 6(3^x) - 27 = 0$
 let $u = 3^x$

$\therefore u^2 - 6u - 27 = 0$
 $(u - 9)(u + 3) = 0$

$\therefore u = 9$ or $u = -3$

$3^x = 9$ $3^x = -3$

$x = 2$ no solution

$\therefore x = 2$

$$e) y = kx^3 + 2$$

$$m = \tan 60^\circ = \sqrt{3}$$

$$\text{at } x=2, m=\sqrt{3}$$

$$y = kx^3 + 2$$

$$\frac{dy}{dx} = 3kx^2$$

$$\frac{dy}{dx} = m$$

$$\therefore \sqrt{3} = 3kx^2 \text{ at } x=2.$$

$$\sqrt{3} = 3k \times 2^2$$

$$\sqrt{3} = 12k$$

$$k = \frac{\sqrt{3}}{12}$$